

trigger mechanism

To enable this protection function, the three object items mentioned above must be correctly specified. When the motor is running above the rated current, the drive detects and calculates the time integral of I^2 . When the integral value is greater than the product of the set maximum current $2000_h : 0A_h$ (0066_h) and the maximum current duration $2000_h : 0B_h$ (0067_h), the motor reports an overload warning and restricts the torque output to the rated value. If these conditions are not met, the $I^2 t$ overload protection function will not be triggered.

6.5 Brake Settings

When the torque and speed of the motor are in the opposite direction, the energy is transferred back from the motor end to the driver of the servo motor, which makes the bus voltage value rise, and when it rises to the braking point, the energy needs to be consumed through the braking resistor, otherwise the driver of the servo motor will be damaged. The braking resistor of the integrated servo motor is divided into two types: built-in and external ($2002_h : 13_h$ (00C3_h) settings, 0~internal (factory default), 1~external, valid for power-on again, factory default 55V trigger), but cannot be used at the same time. The built-in braking resistor specifications are shown in the table below:

Table 6- 15

Capacity	Servo Motor Model	Built-in braking resistor specifications	Handling power
100W	PMM4010B	50Ω, 10W	3W
200W	PMM6020B	15Ω, 25W	8W
400W	PMM6040B	15Ω, 25W	8W
750W	PMM8075B	15Ω, 35W	11W

6.5.1 Torque without external loads

Motor to do back and forth reciprocating action, brake kinetic energy will be converted into electrical energy back to the bus capacitor, to be the bus voltage over the braking voltage $2001_h : 0F_h$ (0093_h), the braking resistor will consume the excess feed-back energy. The following table shows the braking energy E generated by the servomotor from 3000rpm to standstill :

Table 6- 16

power (output)	Servo motor model	Rotor inertia (10^{-4} kgm^2)	Braking energy E (J)
200W	PMM6020B	0.19	0.94
400W	PMM6040B	0.3	1.48
750W	PMM8075B	1.2	5.94

The selection of braking resistor power P is related to the period of reciprocating motion T (unit S), motor speed and the ratio of inertia of the load to the motor N. The specific calculation formula is as follows:

$$P = \frac{2 \times (N+1) \times E}{T}$$

The specific speed profile of the servo motor from 3000rpm at no load to standstill is as follows:

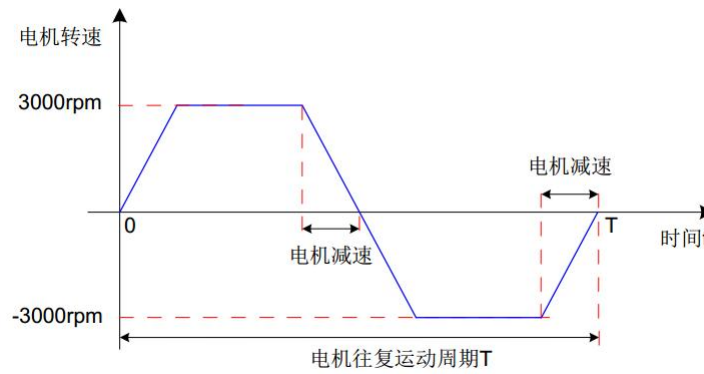


Figure 6- 8

Taking 200W capacity as an example, assuming the reciprocating motion period $T=2s$, the maximum speed of 3000rpm, and the load inertia is 10 times of the motor inertia, the braking resistor power is required:

$$P_a = \frac{2 \times (N+1) \times E}{T} = \frac{2 \times (10+1) \times 0.94}{2} = 10.34W$$

P_a is less than the power 12.5W that can be handled by the built-in braking resistor, so the requirement can be met by using the built-in braking resistor.

If the load inertia in the above assumptions is changed from 10 times to 30 times, and other conditions remain unchanged, the braking resistor power P_b is required:

$$P_b = \frac{2 \times (N+1) \times E}{T} = \frac{2 \times (30+1) \times 0.94}{2} = 29.14W$$

P_b is greater than the power that can be handled by the built-in braking resistor, 12.5 W. Therefore, an external braking resistor is required. When using an external braking resistor, it is necessary to use 30% of the rated power, i.e., $P_b / (1-70\%) = 97.13W$.

6.5.2 With external load torque

The direction of rotation of the motor is the same as the direction of rotation, and the motor outputs energy to the outside. However, there are some special occasions when the motor torque output is opposite to the direction of rotation, at which time the motor performs negative work, and the external energy is fed back to the driver through the motor to generate electrical energy.

Taking the 400W (rated torque 1.27Nm) capacity as an example, when the external load torque is 60% of the rated torque and the rotation speed reaches 1,000rpm, the power fed back to the servo motor driver is calculated as follows:

$$P = \frac{2\pi \times n \times M}{60} \times 60\% = \frac{2\pi \times 1000 \times 1.27}{60} \times 60\% = 79.76W$$

Considering that the braking resistor needs to be used at 30% of the rated value, the power of the external braking resistor $P_{\text{实际}}$ is $79.76 / (1-70\%) = 265.85W$, with a resistance value of 15Ω.

Note: n is rotational speed (rpm), M is torque (N.m)

Note: This product is not suitable for use as a generator unless the DC side power supply allows energy return.

6.6 Parameter Saving and Restoration

Parameter management is written according to the CiA301 standard, 1010_h (0026_h ~003A_h) is the save parameter function object, 1011_h (003C_h ~0050_h) is the read parameter function object.

6.6.1 Parameter Saving

In 1010_h (0026_h ~003A_h), the value of 1 indicates that the corresponding save operation is supported, 0x65766173 is the save parameter command, write the save parameter command word to 1010_h (0026_h ~003A_h), return 1 for the success of the save parameter, and return non-1 for the unsuccessful.

For example: 1010_h : 01_h (0026_h) write 0x65766173 for saving the parameters to the user area, 1010_h : 0B_h (003A_h) write 0x65766173 for saving the NiMotion parameters.

6.6.2 Parameter Recovery

In the sub-index of 1011_h, a value of 1 indicates that the corresponding recovery operation is supported. 0x64616F6C is recovery parameter commands, write the recovery parameter command word to the corresponding sub-index of 1011_h, and a return of 1 indicates that the recovery parameter is successful, and a return of non-1 indicates that it is unsuccessful.

For example: 1011_h : 01_h (003C_h) writes 0x64616F6C for restoring the factory area parameters to the user area.

6.7 Position Recovery Function

The motor is able to regain its absolute position after being powered down and then powered back up.

By setting the position recovery method 2005_h : 18_h (012A_h) to a value of 1, it applies to single-turn round-trip motions in which the motor is in the negative direction with respect to the zero point and the return to zero point requires positive operation; and in which the motor is in the positive direction with respect to the zero point and the return to zero point requires negative operation.

Parameter 2005_h : 18_h (012A_h) Position recovery method, when the value is 0 it is multi-turn absolute value recovery, when the value is 1 it is absolute position recovery within a single turn;

For single-turn position recovery, the parameter 2005_h : 1D_h (012F_h) can be set to change the direction of return to zero by blocking the position.

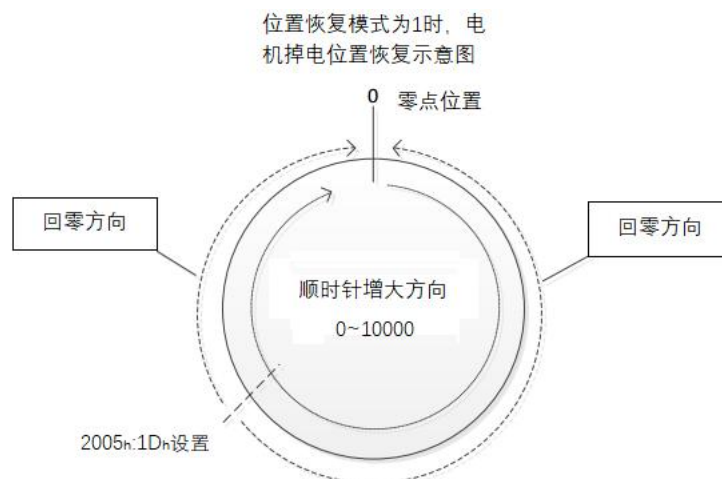


Figure 6- 9

When the value of the position recovery mode 2005_h:18_h (012A_h) is 0, which applies in multi-turn motion trajectories, the motor is able to recover the current absolute position.

6.8 Absolute Encoders

The integrated motor detects the battery voltage once per power-up and updates it in parameter 200F-0B_h. When the integrated motor detects that the battery voltage drops below a specified value during operation, the integrated motor issues a warning, fault message.

Table - 617 Related Information

Battery Voltage	warning message	Alarm Recovery	Battery Replacement Method
$2.6V > DC > 2.4V$	0x7302	Multi-turn encoder counting errors may occur; battery replacement recommended	1. The integrated motor is powered up and in a non-operational state; 2. Replace the batteries in the battery mounting box; 3. The alarm is automatically disarmed.
$DC \leq 2.4V$	0x7301	Please replace the battery immediately	
$DC \geq 2.6V$	0x7309	Contact technical support to recalibrate the multiturn encoder	-

6.9 STO Safety Functions

Safe Torque Off (STO) is a safety function. STO function is triggered by the external hardware terminals STO1 and STO2 of the drive. After the STO function is triggered, the drive will turn off the input signal of the drive chip, and at the same time turn off the drive control signal to cut off the input current of the motor, preventing the drive from generating torque in the shaft end of the motor, so as to make the motor stop running.

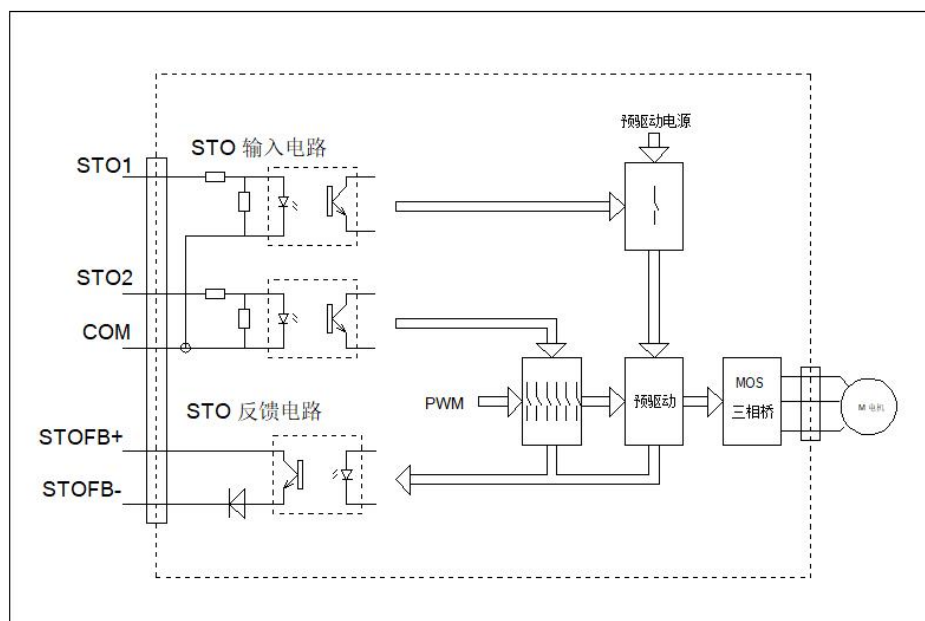


Figure 6- 10 STO function wiring diagram

STO1/STO2 must be connected to the +24VDC signals at the same time so that they are at a high level to

enable normal operation of the driver. If either or both of them are placed low at the same time, then the PWM signal will be blocked.

Table 6-18 STO Function Table

STO1 Input	STO2 Input	Motor Input Current
H	H	normalcy
L	H	interdict
H	L	interdict
L	L	interdict

Table 6-19 STO (Safety Torque) Related Descriptions

Event	Clarification
Definition	Only cuts power to the motor; it does not cut power to the drive.
Description	The STO function allows the servomotor to safely enter a torque-free state and prevents accidental startup. When the STO function is activated, if the motor is running, the drive immediately blocks the output and the servomotor enters a free stop state. If the servo motor has a holding brake, the brake closes immediately and the motor stops immediately.
STO Enabled State	When at least one of the input signals of STO_1 or STO_2 becomes low, the STO enable state is entered. The power drive power supply and drive signal are cut off, and the servo enable command is invalid.
Non-Safe State	When switching from the STO enable state to the normal operation state, it will first enter an unsafe state. In this state, the power drive part will no longer work, and the STO_1/STO_2 input signals and servo enable instructions will not take effect. The servo will start self-checking. If the self-check is passed, the non-safe state will be exited and switched to the normal operating state. If the self-check fails, it will remain in a non-safe state and alarm for the detected faults. After clearing the faults, the self-check will be conducted again until it passes and the non-safe state will be exited. 2.After the servo motor is powered on, it will first enter an unsafe state. In this state, the power drive part will no longer work, and the STO_1/STO_2 input signals and servo enable instructions will not take effect. The servo will start a self-check. If the self-check is passed, it will exit the unsafe state and switch to the normal operation state or the STO enable state. If the self-check fails, it will remain in a non-safe state and alarm for the detected faults. After clearing the faults, the self-check will be conducted again until it passes and the non-safe state will be exited.

6.10 Communication Watchdog Function

The communication watchdog function is designed to enable the servo motor to promptly alarm and stop rotating when communication is disconnected to prevent accidents.

When parameter H0249 = 0, this function is turned off. When H0249 $\neq$ 0, this function is activated, and the dog feeding time is the parameter value of H0249. If the integrated motor does not receive the dog feeding command within the set time, the motor will report a communication failure 0x7501.

For example: Suppose the ID of the motor on the modbus bus is 1. Enable the watchdog function and set the watchdog time of the integrated motor to 100ms, then turn off the watchdog function.

Serial number	Function	Message	Note
1	Modify the watchdog time of the target motor by 100ms	01 10 02 49 00 01 02 00 64	Effective immediately
2	Repeat feeding the dog	01 10 02 49 00 01 02 00 64	No alarm
3	Turn off the watchdog function	01 10 02 49 00 01 02 00 00	
4	Save parameters	01 10 00 26 00 02 04 65 76 61 73	
5	Restart takes effect		The watchdog function is turned off.

7 Adjustments

7.1 Overview

In order to maximize the performance of the integrated servo motor so that the device can respond quickly and accurately to commands from the host computer or internal settings, the gain must be reasonably adjusted according to the characteristics of the drive load. Gain includes speed loop gain, position loop gain, speed loop integration time, filtering, etc. They affect each other, so when adjusting the gain parameters, the balance between each parameter must be considered.

Note: Before proceeding with the gain adjustment, it is recommended to carry out a spot test run to confirm that the motor can operate normally; the adjustment process may be accompanied by vibration of the motor, so please pay full attention to safety.

7.2 Manual Gain Adjustment

7.2.1 Relevant Object

Table 7 - 1

Object Index	Description
2008 _h : 01 _h (0178) _h	Velocity ring gain (unit 0.1Hz), in the case of no noise and vibration, increase this parameter, can speed up the positioning time, bring better speed stability and followability; generate noise, then reduce the parameter setting value;
2008 _h : 02 _h (0179) _h	Speed ring integral time constant (unit 0.01ms), reduce the set value can strengthen the integral effect, speed up the positioning time, but the set value is too small easy to cause mechanical vibration.
2008 _h : 03 _h (017A) _h	Position loop gain, increase this parameter to speed up the positioning time and improve the ability to resist external disturbances when the motor is stationary. Too high a setting may lead to system instability and oscillation.
2008 _h : 0F _h (0186) _h	Speed feed-forward filtering time constant (in 0.01ms)
2008 _h : 10 _h (0187) _h	Velocity Feed Forward Gain
2008 _h : 11 _h (0188) _h	Torque feed-forward filtering time constant (in 0.01ms)
2008 _h : 12 _h (0189) _h	Torque Feed Forward Gain
2007 _h : 06 _h (0163) _h	Torque command filtering time constant to eliminate high frequency noise and suppress mechanical resonance. Increase 2008 _h : 01 _h (0178 _h) When vibration occurs, it can be suppressed by adjusting 2007 _h : 06 _h (0163 _h). Too large a setting will result in reduced response of the current loop

7.2.2 Basic Gain Adjustment Instructions

The integrated servo motor consists of three control loops, current loop, speed loop and position loop in order from the inside out, with the current loop having the highest response frequency. The factory default current loop gain parameter of the servo motor has ensured sufficient responsiveness and generally does not need to be adjusted, so the only things that need to be adjusted are the

position loop gain, the speed loop gain and the speed loop integration time constant. In order to ensure the stability of the system, improve the position loop gain at the same time, need to improve the speed loop gain.

The basic gain adjustment is as follows:



Figure 7- 1 Velocity loop gain



Figure 7- 2 Velocity loop integration time constant



Figure 7- 3 Position loop gain



Figure 7- 4 Torque command filtering time constant

7.2.3 Parameter tuning based on rigid tables

Based on the rigidity table parameter adjustment refers to the rigidity table provided by the manufacturer (the following table) to set the PID parameters corresponding to different rigidity levels, to realize the purpose of quickly adjusting the stability of the integrated motor.

Table 7-2 Parameters updated by rigid table tuning mode

rigidity	2008-03 _h (017A _h) Position Gain (Hz)	2008-01 _h (0178 _h) Speed loop gain (0.1Hz)	2008-02 _h (0179 _h) Speed Integration Time Constant (0.01ms)
1	2	15	37000
2	3	20	28000
3	3	25	22000
4	4	30	19000
5	5	35	16000

6	6	45	12000
7	8	60	9000
8	10	75	7000
9	12	90	6000
10	14	110	5000
11	18	140	4000
12	32	180	3100
13	39	220	2500
14	48	270	2100
15	63	350	1600
16	72	400	1400
17	90	500	1200
18	108	600	1100
19	135	750	900
20	162	900	800
21	206	1150	700
22	251	1400	600
23	305	1700	500
24	377	2100	400
25	449	2500	400
26	500	2800	350
27	560	3100	300
28	610	3400	300
29	660	3700	250
30	720	4000	250
31	810	4500	200
32	900	5000	200

Under no-load conditions, it is recommended that the PID parameters be set to those corresponding to rigidity class 12 or 13.

When it is necessary to adjust the parameter, it is recommended to prioritize the PID parameter with low rigidity level, and then increase the rigidity level step by step until it meets the usage requirements.

The PID parameters can be fine-tuned when the desired effect is not achieved by following the parameter settings in the rigidity table. The effect is optimized by more detailed adjustments.

Note: Before applying the rigidity table adjustment parameter, please confirm with the technical support staff whether the purchased integrated motor supports this function.

7.2.4 Feedforward Gain Adjustment Instructions

Speed Feed Forward

Velocity feedforward improves the velocity command response and reduces the position deviation at fixed velocity, and is mainly used in the fully closed-loop position control mode. The basic adjustment method is as follows:

Set $2008_h : 0F_h$ (0186_h) as a fixed value; then, gradually increase the setting value of $2008_h : 10_h$

(0187_h) from 0 until the speed feedforward achieves an effect at a certain setting value. When adjusting, the 2008_h : 0F_h (0186_h) and 2008_h : 10_h (0187_h) settings should be adjusted repeatedly to find a well-balanced setting.

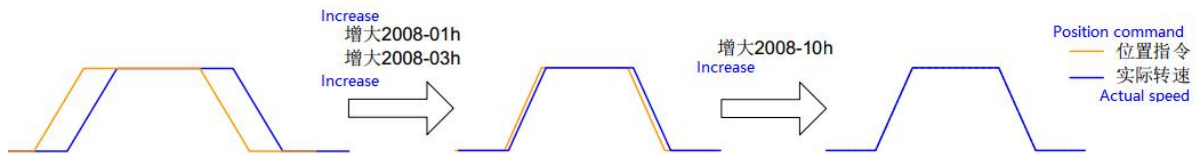


Figure 7- 5 Velocity feedforward

Torque Feed Forward

Position control mode, using torque feed-forward, can improve the torque command response and reduce the position deviation at fixed acceleration and deceleration; speed control mode, using torque feed-forward, can improve the torque command response and reduce the speed deviation at fixed speed.

The basic adjustment method is as follows:

Set 2008_h : 11_h (0188_h) to a fixed value; then, gradually increase the 2008_h : 12_h (0189_h) setting from 0 until the speed feedforward is effective at a certain setting. When adjusting, the 2008_h : 11_h (0188_h) and 2008_h : 12_h (0189_h) settings should be adjusted repeatedly to find a well-balanced setting.

7.2.5 Gain Switching Adjustment Instructions

The gain switching function of the motor is triggered by the external DI or internal state, when the switching condition is satisfied, the motor switches to the group 1 or 2 gain according to the validity of the relevant parameter settings to meet the actual use requirements, the specific role is as follows:

- The motor running state is switched to a higher gain for better command tracking performance;
- The motor standstill is switched to a higher gain to shorten the positioning time;
- The motor quiescent (enabled) state is switched to a lower gain to suppress vibration;
- Different gain settings can be switched by external signals depending on the load equipment;

Gain Switching Flowchart

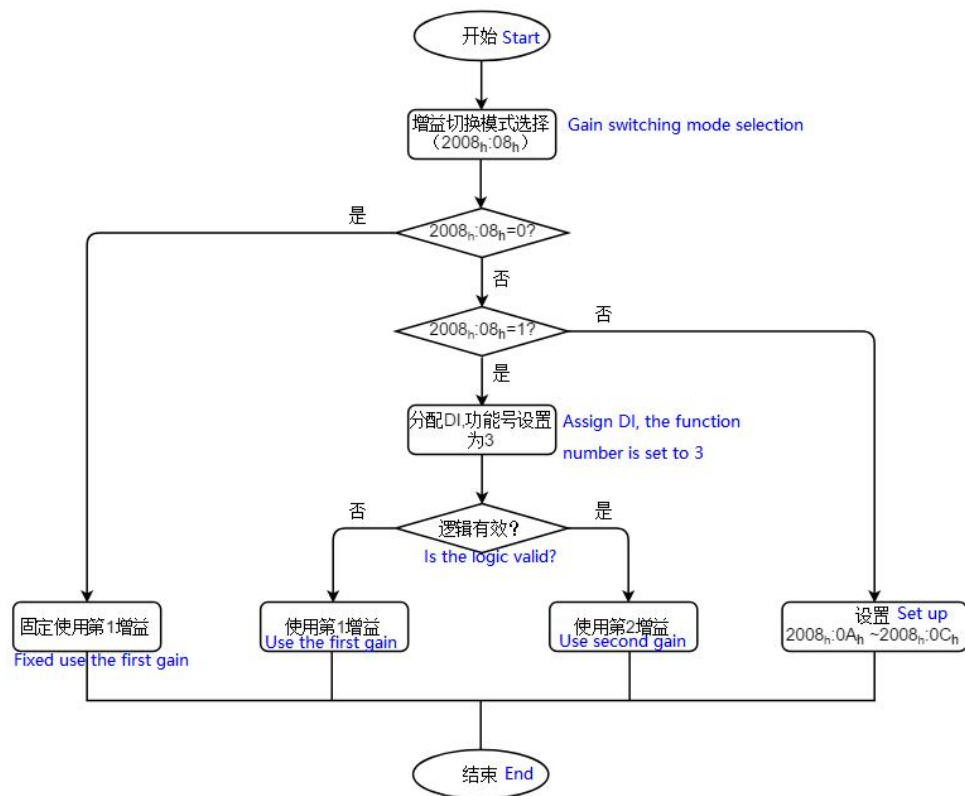


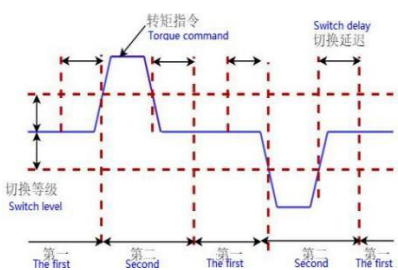
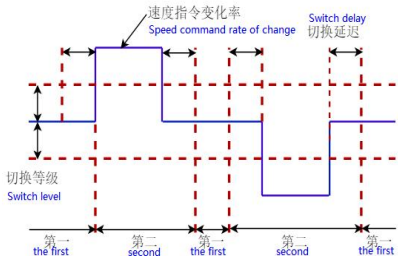
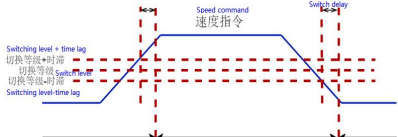
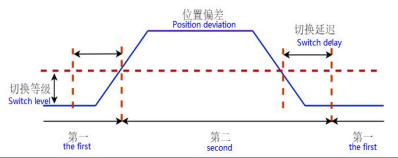
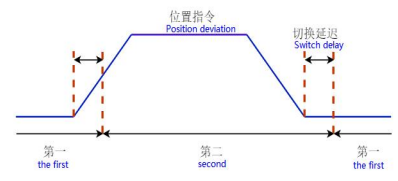
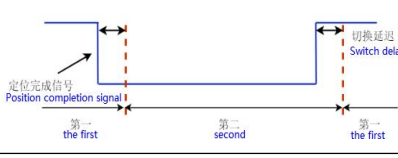
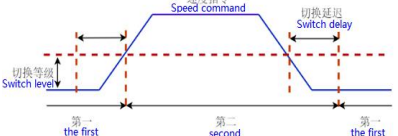
Figure 7- 6

Gain switching conditions

There are 10 modes for switching between the first gain (2008_h:01_h (0178_h) ~ 2008_h:03_h (017A_h)) and the second gain (2008_h:04_h (017B_h) ~ 2008_h:06_h (017D_h)) and there are 10 modes of switching between the different modes as shown in the following table for the schematic diagrams and the associated parameters:

Table 7- 1

Gain switching mode			Associated parameters		
2008 _h :08 _h (017F) _h	model name	schema	latency 2008 _h :0A _h (0181) _h	Toggle Level 2008 _h :0B _h (0182) _h	time lag of switching 2008 _h :0C _h (0183) _h
0	Fixed use of 1st gain	-	null	null	null
1	Switchi ng with extern al DI	-	null	null	null

Gain switching mode			Associated parameters		
2	Torque command		validity (ms)	validity (%)	validity (%)
3	speed command	Internally validated speed command value (606B) _h 25ms constant fixed using gain 1, change using gain 2	validity (ms)	null	null
4	Speed command and rate of change		validity (ms)	validity (rpm/ms)	validity (rpm/ms)
5	speed command		validity (ms)	validity (rpm)	validity (rpm)
6	positional deviation		validity (ms)	validity (Encoder unit)	validity (Encoder unit)
7	position command		validity (ms)	null	null
8	Positioning complete.		validity (ms)	null	null
9	Actual speed		efficiently (ms)	validity (rpm)	validity (rpm)

7.3 Vibration Suppression

7.3.1 Mechanical Resonance Suppression

Servo motors may resonate near the mechanical resonance frequency while the basic gain is increased due to the inherent resonance characteristics of the mechanical system during actual use, which affects the performance of the servo motor. The trap can achieve the purpose of suppressing the resonance by reducing the gain at a specific frequency.

The integrated servo motor has 4 sets of manual traps, each set of traps has 3 parameters, which are trap frequency, trap width and depth, and each parameter is set manually by the user, as shown in the table below:

Table 7- 2

name (of a thing)	Manual Trap			
	1	2	3	4
Frequency (Hz)	2009 _h :0D _h (019E) _h	2009 _h :10 _h (01A1) _h	2009 _h :13 _h (01A4) _h	2009 _h :16 _h (01A7) _h
Trap width (Hz)	2009 _h :0E _h (019F) _h	2009 _h :11 _h (01A2) _h	2009 _h :14 _h (01A5) _h	2009 _h :17 _h (01A8) _h
Depth (%)	2009 _h :0F _h (01A0) _h	2009 _h :12 _h (01A3) _h	2009 _h :15 _h (01A6) _h	2009 _h :18 _h (01A9) _h

Note: When the depth level is 0, the trap is invalid.

When using manual traps, you need to set the trap frequency to the actual center resonance frequency, and enter the width level (center resonance frequency range) and depth level of the trap, if the resonance is suppressed, it means that the traps are effective.

Trap Depth

At trap depth level 100, the input is completely suppressed at the center frequency; at trap depth level 0, the trap is ineffective. Therefore, the larger the trap depth level setting, the deeper the trap depth and the stronger the suppression of mechanical resonance, but it may lead to system instability and care should be taken when using it.

The principle of trap suppression is shown below:

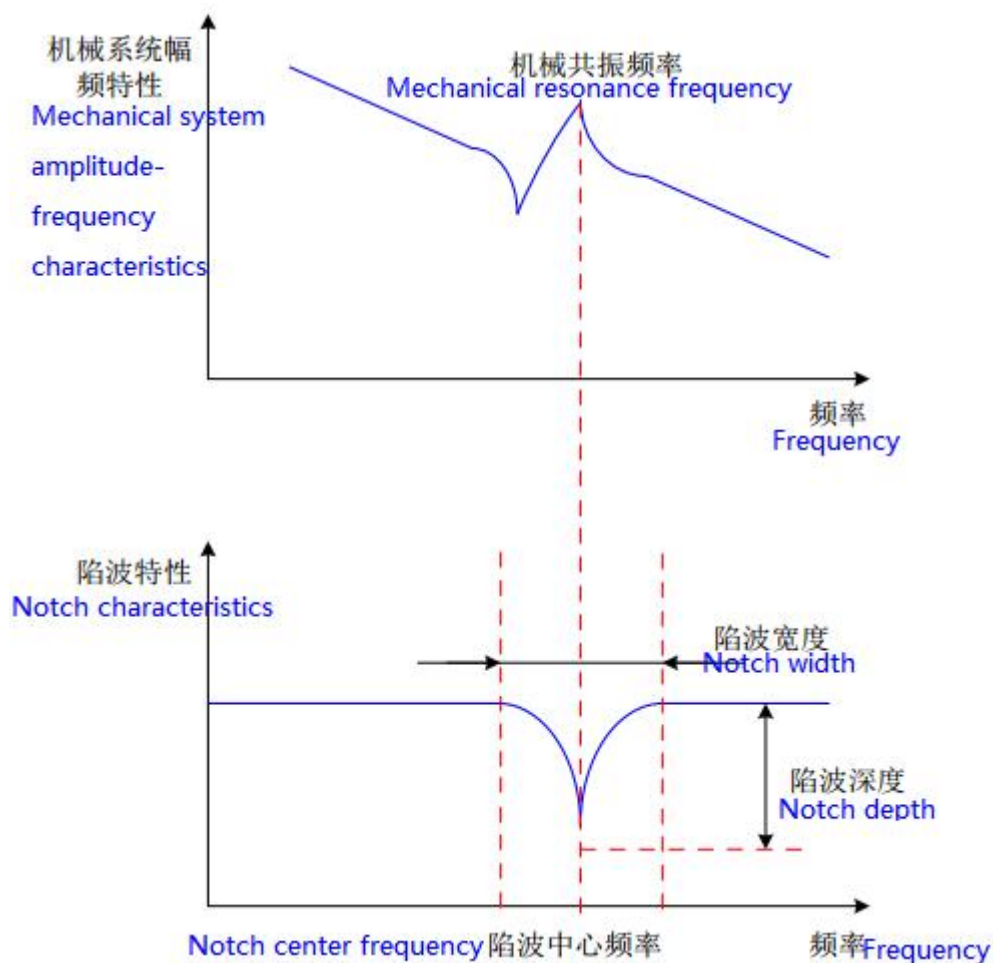


Figure 7- 7

7.3.2 Low Frequency Vibration Suppression

Due to the inherent characteristics of the dual inertia system during position control, end vibration is likely to occur during emergency stops, affecting the positioning effect, and the frequency of this vibration is generally within 100Hz, which can be reduced by the low-frequency vibration suppression function.

relevant object

Table 7- 3

object index	Description
2009 _h :05 _h (0196) _h	Low frequency resonance suppression mode selection, 0-off, 1-on
2009 _h :1D _h (01AE) _h	Low-frequency vibration frequency (unit 0.1Hz), set the center frequency of low-frequency vibration suppression, limit range 10~1000
2009 _h :1E _h (01AF) _h	Vibration suppression width level (unit %), set the effective frequency range of vibration suppression, 0 represents only suppression of vibration at the center frequency; set value is

object index	Description
	adjusted large, can increase the frequency range of low-frequency vibration suppression, but it will lead to longer positioning time; set value is too small, can not complete the suppression of vibration in the load vibration frequency will be changed occasions (such as belt loads), according to the actual situation of debugging settings. Limit range 0~100
2009 _h :23 _h (01B5) _h	Damping ratio coefficient, set the system vibration damping ratio coefficient, the value of the default is 5, generally do not modify, limit the range of 0 ~ 50

Setting low-frequency vibration suppression parameters

- Turn on the low-frequency vibration suppression mode, 2009_h:05_h Set 1;
Send: 01 10 01 96 00 01 02 00 01
- The NiMStudio host computer software was used to capture the waveforms when the motor was in positional localization and to automatically calculate the frequency of the low-frequency resonance after the motion was stopped;

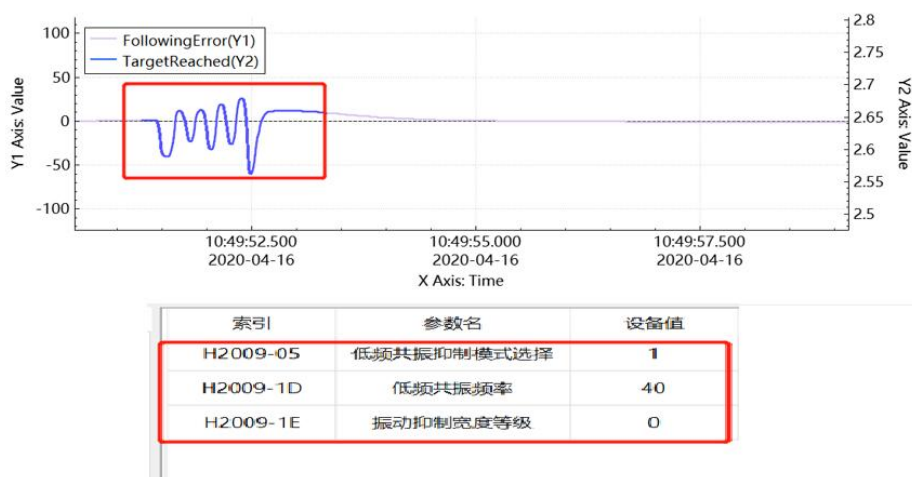


Figure 7- 8

- Set the low-frequency vibration frequency and vibration suppression width level, etc. according to the low-frequency vibration frequency recognized by the NiMStudio PC software;
- Observe the effect of positional positioning of the motor after using the low-frequency vibration suppression function. If it is within the acceptable range, it means that the low-frequency vibration suppression is effective; if it is not within the acceptable range, re-set the above two steps or manually set the low-frequency vibration frequency and the vibration suppression width level.



Figure 7- 9

8 Fault Management

8.1 Alarm Code

When the integrated servo motor has a warning or fault, the motor will take the initiative to alarm (Indicator light - yellow blinking, 0.5s blinking frequency means warning, 0.25s blinking frequency means fault; Indicator light - green 0.5s blinking means communication alarm, constant light means communication is normal), and perform the corresponding action. The corresponding alarm code will be stored in objects 1003_h : 01_h (0002_h)~1003_h : 10_h (0020_h), and object 1003_h : 00_h (0001_h) is the current number of faults.

There are 16 alarm queues in object 1003_h (0002_h~0020_h) that follow the FIFO rule. In the case that all the historical alarm stores are occupied, generating a new alarm will delete the earliest error that occurs, and the previous errors will move down in order. The specific alarm codes are shown in the following table:

Table 8- 1 List of alarm codes

Alarm Code	Alarm content	Default Alarm Type	Default Fault Response Code	Self-reset or not	037F _h (error code)
0x2300	Motor overcurrent	malfunctions	0	No	0x2300
0x4012311	Motor overload	warnings	4	Yes	0x2311
0x3002312	motor stalling	Pause and lock the shaft	3	No	0x2312
0x13210	Power overvoltage	malfunctions	0	Yes	0x3210
0x1013220	Power supply undervoltage	malfunctions	1	Yes	0x3220
0x14210	Over Temperature Alarm	malfunctions	0	Yes	0x4210
0x14220	Low temperature alarm	malfunctions	0	Yes	0x4220
0x5080	Drive failure*	malfunctions	0	No	0x5080
0x5540	Flash operation malfunction*	malfunctions	0	No	0x5540
0x5541	Flash initialization failure*	malfunctions	0	No	0x5541
0x4005542	Flash Checksum Error Warning*	warnings	4	No	0x5542
0x5543	Flash user area without parameters*	malfunctions	0	No	0x5543
0x4005544	Abnormal power-down position storage	warnings	4	No	0x5544
0x4005545	Power-down save data not stored	warnings	4	No	0x5545
0x6000	Hardware initialization	malfunctions	0	No	0x6000

Alarm Code	Alarm content	Default Alarm Type	Default Fault Response Code	Self-reset or not	037F _h (error code)
	failure*				
0x16320	Parameter setting error	malfunctions	0	Yes	0x6320
0x6321	Registration failures*	malfunctions	0	No	0x6321
0x7301	Encoder Battery Low Voltage Failure	malfunctions	0	No	0x7301
0x4017302	Encoder battery undervoltage alarm	warnings	4	Yes	0x7302
0x1017304	Encoder multiturn count error	warnings	1	Yes	0x7304
0x7305	Z-pulse fault*	malfunctions	0	No	0x7305
0x7306	Encoder malfunction*	malfunctions	0	No	0x7306
0x4017307	Encoder warning*	warnings	4	Yes	0x7307
0x17310	speeding	malfunctions	0	Yes	0x7310
0x3017501	Illegal Function Codes in Modbus Communication	warnings	3	Yes	0x7501
0x4017502	Illegal addresses in Modbus communication	warnings	4	Yes	0x7502
0x4017503	Illegal data values in Modbus communication	warnings	4	Yes	0x7503
0x4017505	Confirmation in Modbus communication	warnings	4	Yes	0x7505
0x4017506	Slave device busy in Modbus communication	warnings	4	Yes	0x7506
0x401750C	Synchronization message in Modbus communication requesting data data is greater than the total mapped data	warnings	4	Yes	0x750C
0x401750D	The number of synchronization messages requesting data in Modbus communication does not correspond to the	warnings	4	Yes	0x750D

Alarm Code	Alarm content	Default Alarm Type	Default Fault Response Code	Self-reset or not	037F _h (error code)
	mapping data equally.				
0x401750E	Wrong node address for unicast message under synchronization function in Modbus communication	warnings	4	Yes	0x750E
0x4018311	Torque limiting protection	warnings	4	Yes	0x8311
0x8610	Home return timeout	malfunctions	0	No	0x8610
0x8611	The location is terrible.	malfunctions	0	No	0x8611
0x3018613	Software limit error	Pause and lock the shaft	3	No	0x8613
0x3018614	Limit switch error	Pause and lock the shaft	3	No	0x8614
0x4018615	Errors in curve planning calculations	warnings	4	Yes	0x8615
0x18616	Target position overflow	malfunctions	0	No	0x8616
0x5018617	Curve planning parameters too small	neglect	5	Yes	0x8617
0xFF01	Motor Parameter Recognition Fault*	malfunctions	0	No	0xFF01
0x0401FF02	Parameter Save Failure	warnings	4	Yes	0xFF02
0xFF05	STO_1 failed to close	Fault	0	No	0xFF05
0xFF06	STO_1 failed to enable	Fault	0	No	0xFF06
0xFF07	STO_2 failed to close	Fault	0	No	0xFF07
0xFF08	STO_2 failed to enable	Fault	0	No	0xFF08
0xFF09	STO input abnormal fault	Fault	0	No	0xFF09
0xFF0A	STO enabled state	Fault	0	No	0xFF0A
0xFF0B	Non-safe state	Fault	0	No	0xFF0B
0xFF0C	Encoder calibration timeout	Fault	0	No	0xFF0C

Definition of the corresponding bit of the alarm code

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Malfunction Response Code [8]								Self-reset allowed [8]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Error code [16]															

- Definition of Fault Response Codes

- 0- The state machine switches to the fault state (refer to "4.1 CiA402 state machine"), if 605E_h ≥ 0, then follow the fault shutdown mode selected by 605E_h; otherwise: the power tubes are fully turned off, free shutdown.
- 1- The state machine switches to the fault state (refer to "4.1 CiA402 state machine"), if 605E_h ≥ 0, then the fault shutdown mode is selected according to 605E_h (03C1_h); otherwise: the output is shut down according to the slow curve stop.
- 2- The state machine switches to the fault state (refer to "4.1 CiA402 state machine"), if 605E_h ≥ 0, then the fault shutdown mode is selected according to 605E_h (03C1_h); otherwise: the outputs are shut down according to the fast curve stop.
- 3- Pressing the quick curve suspends the locking axis without switching to the fault state and is not affected by object 605E_h (03C1_h).
- 4- Warning and does not change the current operating state, does not switch to faulty state and is not affected by object 605E_h (03C1_h).
- 5- Ignored and logged in 1003_h (0002_h ~ 0020_h), not switched to the fault state and not affected by object 605E_h (03C1_h).
- 6- Ignored but not logged in 1003_h (0002_h ~ 0020_h), not switched to fault state and not affected by object 605E_h (03C1_h).

- Self-reset allowed

- 0- impermissible
- 1- Allowed (reset automatically when warning or fault is removed)

Notes:

- If the default alarm type in the above table (alarm code list) is "Fault", it means that the motor enters a fault state (status word value 0x228) after this alarm is generated.
- The default alarm type for alarm codes, the default fault response code and whether or not to self-reset can be configured via object 200E_h (025B_h ~ 027B_h).
- Alarms with the * symbol cannot be configured via object 200E_h (025B_h ~ 027B_h). The integrated motor needs to be restarted after a drive and hardware initialization failure to operate normally.
- 603F_h (037F_h) contains the last error of the integrated motor, corresponding to the lower 16 bits of the alarm code.

8.2 Fault Action Setting

Alarm code consists of fault code (16 bits), fault response code (8 bits) and self-reset allow (8 bits) totaling 32 bits. Fault code indicates that the motor in the abnormal state of the fault phenomenon, through the fault code to determine the current state of the motor, generally and the object 603F_h (037F_h) (error code) corresponds to 0 indicates that the motor is currently no fault; fault response code indicates that the motor after the failure of the corresponding mode of action and the treatment of the